

LONGUE-POINTE-DE-MINGAN, QC, Canada

WMO: 715120

Lat: 50.271N

Lon: 64.226W

Elev: 36

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -16.9	(c) -11.9	(d) -23.9	(e) 1.5	(f) -14.8	(g) -19.2	(h) 1.9	(i) -10.0	(j) 18.7	(k) 22.9	(l) 16.5	(m) 22.3	(n) 4.3	(o) 340	(p) 0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	70.9	61.8	68.3	60.7	66.0	59.6	64.3	67.9	62.6	65.9	61.2	64.4	6.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.8	85.7	65.8	61.2	80.8	64.3	59.7	76.5	62.8	29.5	67.9	28.3	65.8	27.3	64.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.2	(b) 13.3	(c) 11.7	(e) -26.0	(f) 78.2	(g) 4.7	(h) 4.4	(i) -29.4	(j) 81.4	(k) -32.2	(l) 84.0	(m) -34.9	(n) 86.5	(o) -38.3	(p) 89.7		
(4) 15.2	(5) 13.3	(6) 11.7	(7) -26.0	(8) 78.2	(9) 4.7	(10) 4.4	(11) -29.4	(12) 81.4	(13) -32.2	(14) 84.0	(15) -34.9	(16) 86.5	(17) -38.3	(18) 89.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.9	9.6	11.3	20.4	32.7	41.9	51.1	57.2	56.6	49.0	40.3	29.9	17.0
	DBStd	18.26	11.60	11.44	10.56	6.19	4.90	5.16	3.92	4.76	5.46	5.59	8.15	10.78
	HDD50	6095	1253	1084	916	519	256	47	1	5	82	306	603	1023
	HDD65	10997	1718	1504	1381	969	716	416	243	262	481	766	1053	1488
	CDD50	577	0	0	0	0	5	81	225	210	52	4	0	0
	CDD65	3	0	0	0	0	0	0	1	2	0	0	0	0
	CDH74	19	0	0	0	0	0	1	10	7	1	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
(14) Wind	WSAvg	5.4	5.7	5.9	6.1	6.1	5.2	5.2	4.7	4.5	4.8	5.3	5.5	5.8
(15) Precipitation	PrecAvg	44.0	3.3	2.7	3.5	3.1	3.6	3.7	3.8	3.4	3.9	4.6	4.8	3.9
	PrecMax	50.9	7.6	5.7	7.1	7.4	8.1	6.5	7.4	5.8	7.0	7.8	8.3	10.0
	PrecMin	38.0	0.7	0.9	1.4	0.7	1.4	1.5	1.9	1.8	1.7	1.5	2.0	1.7
	PrecStd	3.9	1.6	1.1	1.7	1.6	1.7	1.6	1.6	1.1	1.2	1.6	1.6	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	41.0	44.9	53.9	64.1	71.7	75.1	74.2	68.6	60.6	50.4	42.9
		MCWB	36.7	38.5	37.2	42.6	52.7	59.1	62.5	63.8	60.0	54.9	49.0	41.5
	2%	DB	34.2	36.5	40.4	48.1	58.3	67.2	71.2	70.5	64.1	55.2	46.2	38.9
		MCWB	33.3	34.0	34.7	40.4	49.5	57.2	61.6	62.8	57.3	50.9	44.9	37.4
	5%	DB	31.7	32.7	37.3	44.6	55.0	64.1	68.5	68.0	61.1	52.0	43.3	35.9
		MCWB	31.1	31.1	33.2	38.2	47.6	55.5	60.8	61.7	56.0	48.7	41.9	34.5
	10%	DB	28.0	28.9	34.7	42.0	52.1	61.4	66.2	65.7	58.7	49.6	41.2	33.1
		MCWB	27.0	27.3	31.7	37.2	46.1	54.4	60.1	60.4	54.7	46.9	39.7	31.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	39.7	39.2	45.0	55.1	62.9	66.6	66.7	62.8	57.2	49.4	42.0
		MCDB	38.2	40.6	42.9	51.5	59.1	67.8	71.2	71.2	65.8	58.7	49.7	42.5
	2%	WB	33.3	34.0	36.2	42.0	51.3	59.8	64.2	64.5	59.6	52.1	45.3	37.8
		MCDB	33.9	35.5	39.0	45.7	55.7	64.2	68.1	68.0	62.0	53.8	45.9	38.5
	5%	WB	30.6	30.7	34.0	39.8	48.9	57.7	62.5	62.9	57.4	49.6	42.3	34.9
		MCDB	31.3	32.2	36.2	43.4	53.2	62.0	66.1	65.9	60.0	51.1	43.0	35.9
	10%	WB	26.7	27.0	32.2	37.7	47.0	55.6	61.1	61.5	55.3	47.4	40.0	32.2
		MCDB	27.5	28.3	34.4	41.1	51.2	59.6	64.6	64.5	57.8	49.0	41.0	33.1
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	20.2	19.7	15.7	17.5	18.1	17.2	17.6	17.7	14.7	13.5	16.4
		MCDBR	17.0	15.9	16.8	20.0	23.5	23.0	22.1	20.2	20.7	17.8	13.0	12.3
		MCWBR	15.9	14.5	12.8	13.0	15.2	13.5	13.3	12.9	14.3	13.6	11.7	11.4
	5% WB	MCDBR	16.9	15.2	15.0	16.8	20.5	18.4	17.5	16.2	17.1	15.0	12.1	12.4
		MCWBR	16.7	14.0	12.4	11.9	14.3	12.4	12.0	11.6	13.3	12.9	11.6	12.1
(40) Clear-Sky Solar Irradiance	taub		0.266	0.270	0.293	0.338	0.362	0.384	0.417	0.381	0.355	0.330	0.291	0.273
	taud		2.361	2.330	2.335	2.334	2.409	2.381	2.334	2.439	2.480	2.516	2.518	2.372
	Ebn at Noon		249	279	291	287	284	277	265	271	267	254	241	227
	Edh at Noon		20	27	32	36	35	36	38	32	28	23	17	17
(44) All-Sky Solar Radiation	RadAvg		334	644	1061	1426	1659	1774	1681	1429	1080	658	345	258
	RadStd		38	53	93	120	100	162	154	73	70	48	32	23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	+1.17	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	+2.23	N/A	+0.59	+0.65	N/A	N/A	N/A	N/A

Nomenclature: See separate page